

AKHIL BAJA

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PROFESSIONAL SUMMARY

Applied AI/ML Systems Engineer with 3+ years of experience building production-grade GenAI, LLM infrastructure, and machine learning systems at **Qualcomm** and **Axis Bank**. Engineered cross-language LLM context engineering pipelines reducing context retrieval latency by **60%** and improving code-generation accuracy by **46%**, and enterprise GenAI assistants serving **1,300+** employees. MSc student in Artificial Intelligence Applications and Innovation at **Imperial College London**.

EDUCATION

Imperial College London – MSc in Artificial Intelligence Applications and Innovation *Sept 2026 – Sept 2027*
London, UK

Indian Institute of Technology Hyderabad (IIT Hyderabad) – B.Tech in Engineering Science *Aug 2019 – May 2023*
CGPA: 8.04 / 10.0

Internship & Placement Cell Coordinator — led corporate outreach recruitment across **100+ tier-1 & engineering firms**.

WORK EXPERIENCE

Qualcomm | Software Engineer (GenAI / Systems) *Dec 2024 – Aug 2026*

- Engineered a cross-language LLM context-engineering pipeline using AST-based dependency extraction across Java, Python, C, and C++, reducing context retrieval latency by **60%** and improving code-generation accuracy by **46%**; received the **Qualcomm Certificate of Recognition**.
- Re-architected a distributed data-harvesting pipeline serving **1,000+** internal engineers, reducing deployment failures by **84%** through asynchronous RabbitMQ queues and containerized Docker/Rancher microservices.
- Slashed static-analysis parser execution latency by **93%** through in-memory dependency caching and consolidated **200+** unit tests into a Linux-compatible test harness, cutting ongoing maintenance overhead by **60%**.

Axis Bank | Data Scientist (AI / ML) *July 2023 – Dec 2024*

- Built and deployed an enterprise RAG assistant using Azure OpenAI, FAISS, and Cloudera ML, serving **1,300+** bank employees and reducing internal support query resolution time by **70%**.
- Designed an Isolation Forest anomaly-detection pipeline for Gold Loans, identifying high-risk branch patterns associated with **£30M+ (INR 300Cr+)** in potential loss exposure and reducing manual audit time by **80%**; received the **BIU Star Award**.
- Automated customer Dynamic Risk Rating using XGBoost and Naive Bayes pipelines, reducing manual processing effort by approximately **40%** while ensuring compliance with RBI regulatory audit standards.

PROJECTS

- AI Job Finder Agent & ATS Tailoring Engine** | GitHub | PyPI | Live Demo
Python, FastAPI, Hugging Face Spaces, Tectonic, Playwright, MCP
Built and published an autonomous CLI agent that searches LinkedIn, Ashby, Greenhouse, and Lever, scores roles against user profiles, dynamically generates tailored one-page resumes, and automates application workflows using Playwright; exposed the pipeline through an MCP server for integration with Cursor and Claude Code.
- AI Study Copilot — Autonomous Lecture Assistant** | GitHub | Live Demo
Python, React, Docker, Telegram Bot API, Hugging Face
Built a cloud-deployed STEM learning assistant supporting lecture transcription, slide OCR, and contextual Q&A through Telegram; implemented a Render reverse proxy for reliable Telegram to HuggingFace inference connectivity.
- Cosmic Ray Segmentation** – *Python, PyTorch*
Enhanced the deepCR model for cosmic-ray detection in astronomical images; evaluated U-Net and implemented residual connections, improving pixel-level segmentation precision by **15%**.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

AI/ML & GenAI: LLMs, RAG, LLM Context Engineering, FAISS, PyTorch, Scikit-learn, XGBoost, Vector Search, Anomaly Detection, Computer Vision (U-Net, DenseNet)

Data & Platforms: PySpark, Azure OpenAI, Hugging Face Spaces, Cloudera ML, PostgreSQL, FastAPI, REST APIs, GCP
Systems, DevOps & Automation: Distributed Systems, Microservices, RabbitMQ, Docker, Rancher, Jenkins, Git, CI/CD, Static Analysis, AST Parsing, Linux, Playwright, Model Context Protocol (MCP)